

Roll Number: Solutions

Name: _____

Quiz 1
EE 3203: Power Electronics

Total Marks: 20

Time: 50 mins

Instructions

1. Write your roll number and name at the top of this question paper.
2. Check whether the question paper has all the pages.
3. The answers are to be filled in the space provided below the questions.
4. Use only pencil for sketching any waveforms/figures.
5. Rough sheet is available at the end of the question paper. Extra papers will not be provided.

Section A

Marks: 5 x 1 = 5

1. Answer the questions below:

- (a) Give an example of a majority carrier power device. (1)

① Power MOSFET

- (b) In power electronic systems, typically the devices are operated as switches. Name the two regions of the output characteristics that they typically operate in, as a switch. (1)

① Cut-off region, Saturation region

- (c) An ac to ac conversion is termed as Cyclo conversion. (1)

- (d) To turn-off a SCR, the anode current should be brought down below holding current and a reverse bias is to be applied for a

① minimum duration/time called as the device turn off time.

- (e) The output ac voltage of a power electronic system was found to have the following components. Compute the total rms voltage of the harmonics. (1)

Frequency Component	rms voltage
Fundamental	105 V
3 rd harmonic	35 V
5 th harmonic	21 V
7 th harmonic	15 V

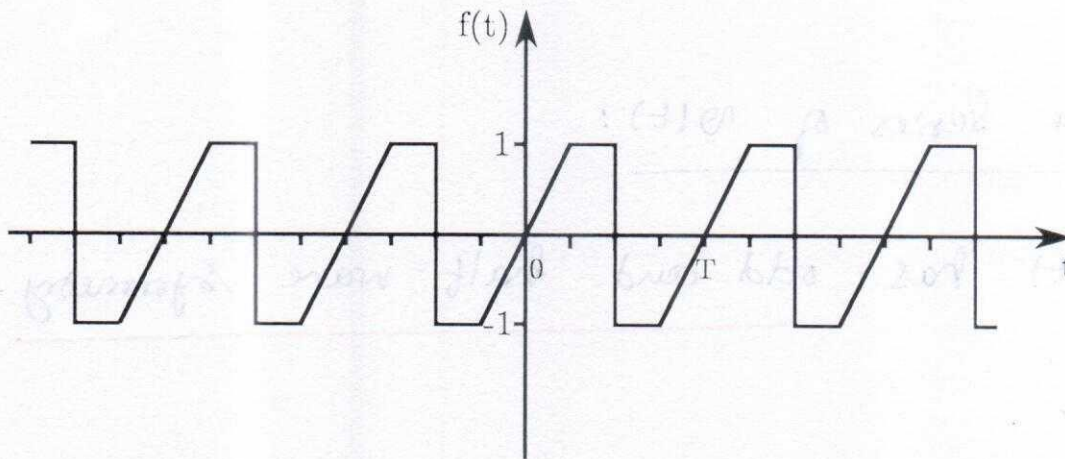
① Total rms voltage of the harmonics = $\sqrt{35^2 + 21^2 + 15^2}$

= 43.49 V.

Section B

Marks: $5 \times 3 = 15$

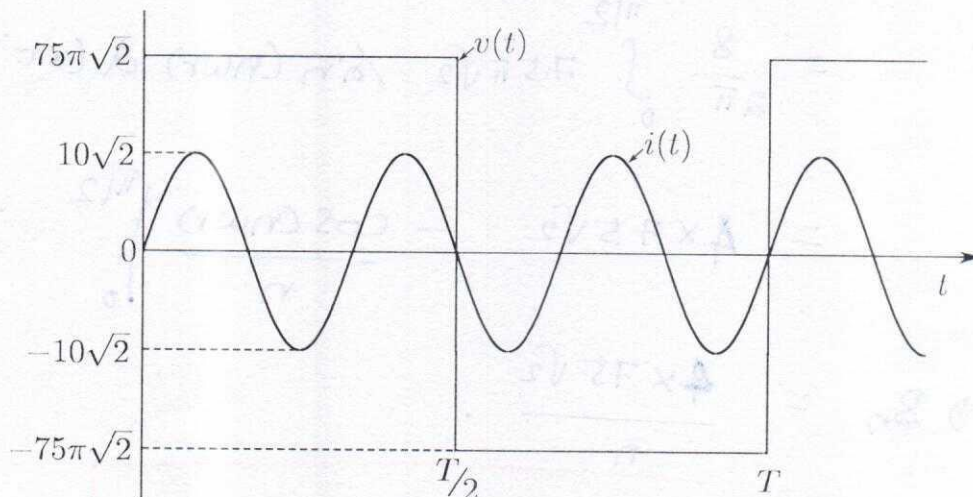
1. (a) List the symmetries, if any, in the following waveform. (5)



①

Symmetries: odd symmetry.

- (b) Assume that the figure below shows the voltage $v(t)$ across and current $i(t)$ through the input terminal of an electrical equipment. Compute the power absorbed by the equipment. The voltage has a square wave nature and the current is a third harmonic sine wave.



Only the third harmonic component of the voltage will interact with the third harmonic current to give an useful power.

Fourier Series of $v(t)$:

$v(t)$ has odd and half-wave symmetry.

So,

$$v(t) = \sum_{n=1,3,5,\dots} B_n \sin(n\omega t)$$

$$B_n = \frac{8}{2\pi} \int_0^{\pi/2} v(t) \sin(n\omega t) d(\omega t)$$

$$= \frac{8}{2\pi} \int_0^{\pi/2} 75\pi\sqrt{2} \sin(n\omega t) d(\omega t)$$

$$= 4 \times 75\sqrt{2} - \frac{\cos(n\omega t)}{n} \Big|_0^{\pi/2}$$

$$\Rightarrow B_n = \frac{4 \times 75\sqrt{2}}{n}$$

$$\Rightarrow v(t) = \sum_{n=1,3,5,\dots} \frac{4 \times 75\sqrt{2}}{n} \sin(n\omega t)$$

③

$$V_3(t) = \frac{4 \times 75 \sqrt{2}}{3} \sin(3\omega t)$$

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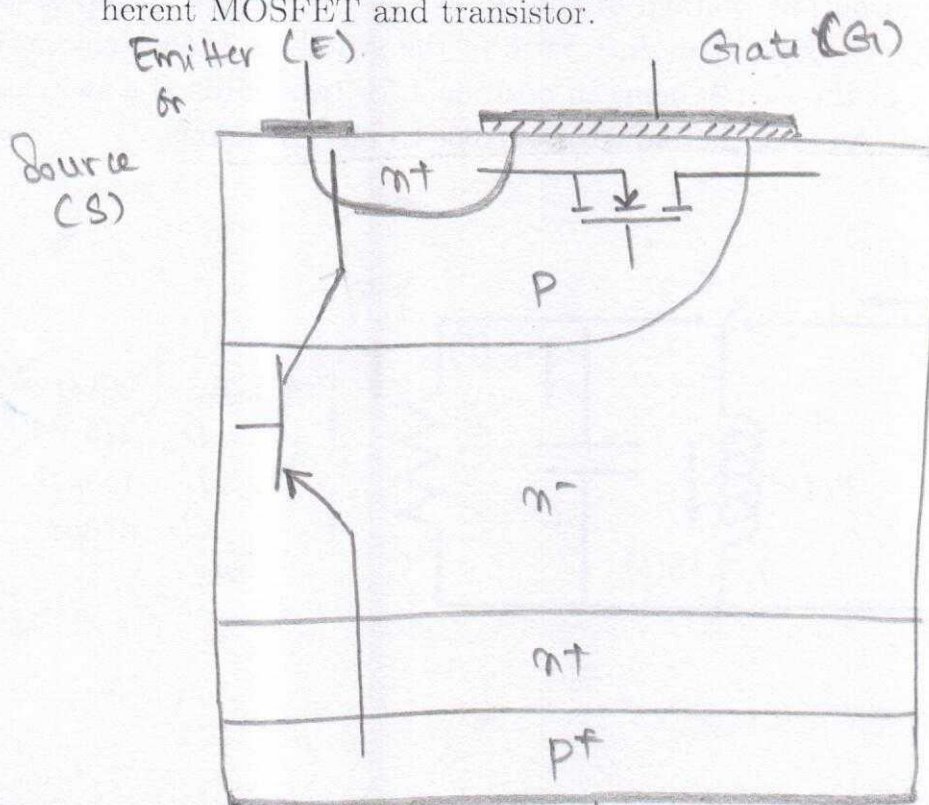
$$i(t) = 10\sqrt{2} \sin(3\omega t)$$

$$\Rightarrow \text{Average power} = P_{\text{avg}} = \frac{4 \times 75}{3} \times 10 \times \cos 0^\circ$$

$$\Rightarrow P_{\text{avg}} = 1000 \text{ W}$$

①

2. (a) Draw the two dimensional structure of an IGBT and show the inherent MOSFET and transistor. (5)



2 Dimensional
Structure of
an IGBT

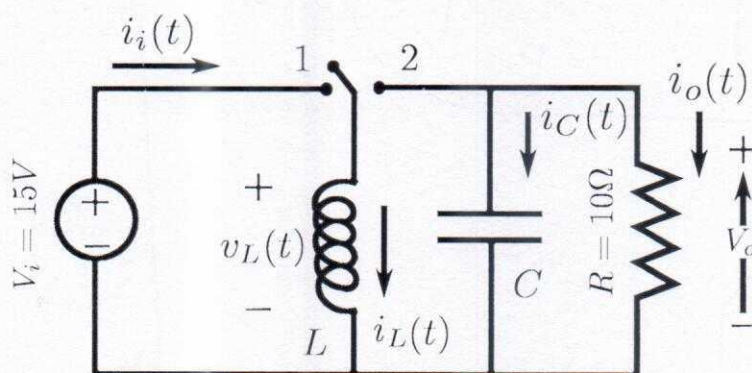
Collector (C)
or
Drain (D)

② 1/2

- (b) Using the structure, explain briefly why the current in an IGBT is typically low in the transistor path.

From the structure of the IGBT, it can be observed that the current can flow through the transistor path or the MOSFET path. Further, the base of the transistor can be observed to be thick. The thick base largely reduces the β (current gain) of the inherent transistor. This causes a lower current in the transistor path.

3. The average output voltage of the buck-boost converter shown in the figure is $V_o = -30V$. The duty ratio of the switch is $2/3$ which can be interpreted as the switch being in position 1 for two-thirds of a switching time period. Assume the inductor current to be continuous.



$$\begin{aligned} f_{sw} &= 20kHz \\ D &= 2/3 \\ L &= 150\mu H \\ C &= 470\mu F \end{aligned}$$

In steady state,

- Assuming the converter to be lossless, compute the average input current I_i of the converter.
- Draw the equivalent circuit of the converter in the two switch positions.
- Using the equivalent circuit, sketch the voltage $v_L(t)$ across the inductor. Consider the output voltage to be a steady dc for sketching the inductor voltage.
- Using the equivalent circuit, voltage $v_L(t)$ and average input current I_i sketch the current $i_L(t)$ through the inductor including the ripple.
- Compute the peak-to-peak current ripple $\Delta i_{L,pk-pk}$ in the inductor.
- Sketch the current $i_C(t)$ through the capacitor. Compute the peak-to-peak voltage ripple $\Delta v_{C,pk-pk}$ in the capacitor.

a. With no losses, Input power = Output power.

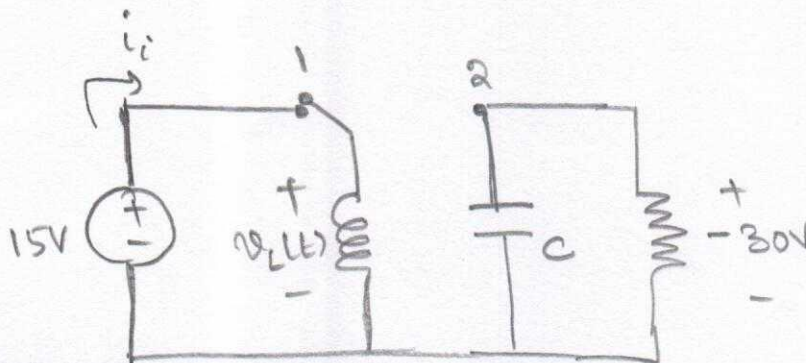
$$V_i \times I_i = V_o \times I_o$$

$$\Rightarrow 15 \times I_i = -30 \times -\frac{30}{10}$$

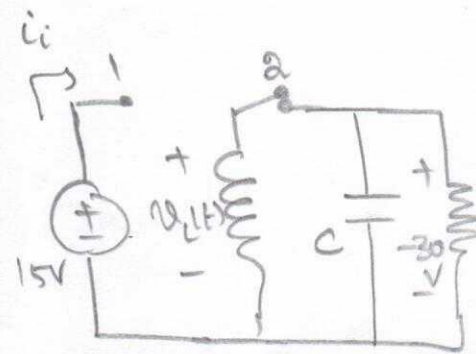
$\Rightarrow$

$$I_i = 6 \text{ A}$$

b.

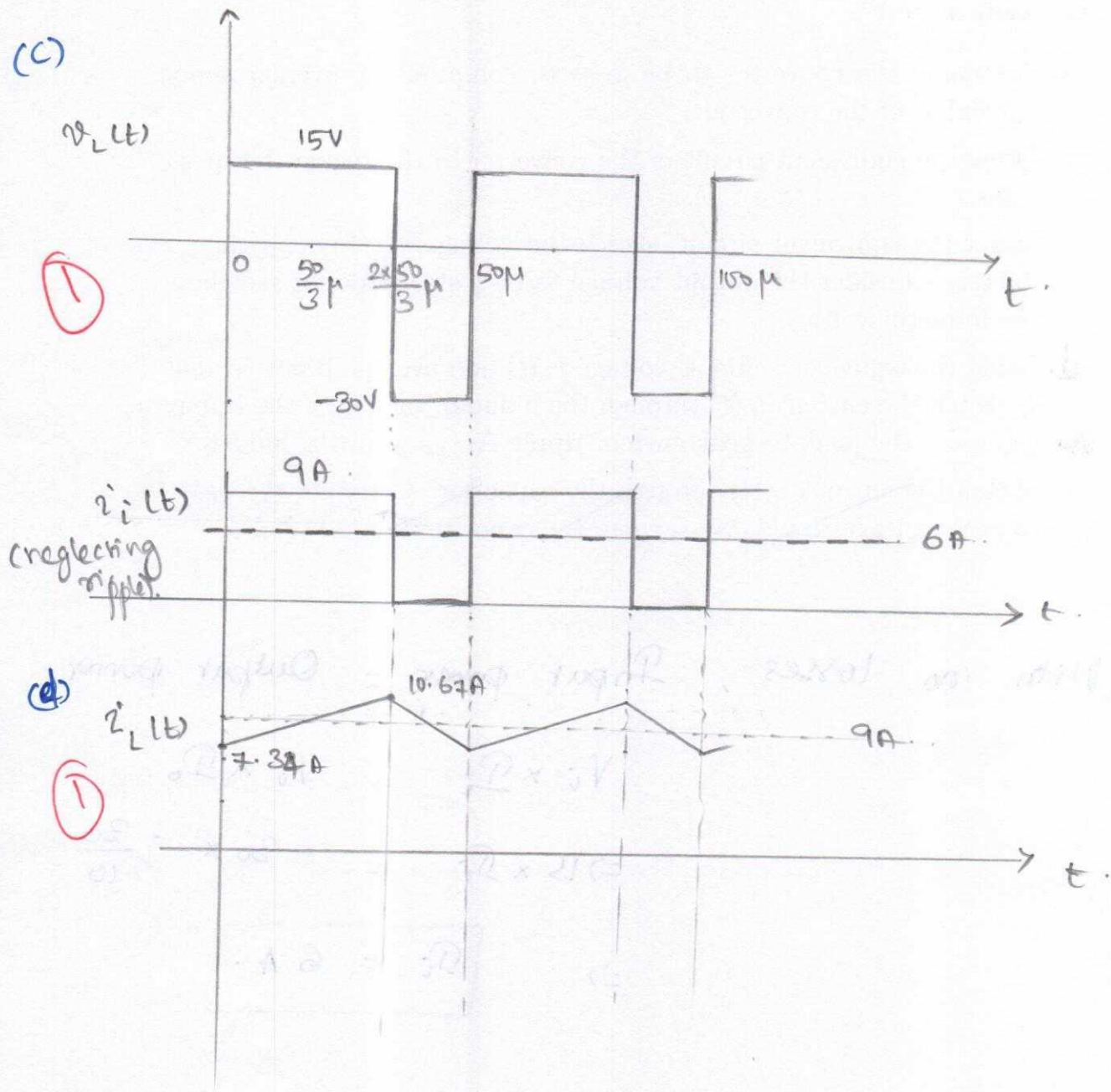


Switch position ①



Switch position ②

(C)



$$v_L = L \frac{di_L}{dt}$$

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d. $\Delta i_{L, pk-pk} = \frac{15}{150 \mu} \times \frac{2}{3} \times 50 \mu.$

$= \frac{10}{3} = 3.33 \text{ A.}$

(P)

Rough Work

$$A = \frac{10}{2} = 5$$